

 SST SCHOOL OF SCIENCE AND TECHNOLOGY, SINGAPORE SCIENCE DEPARTMENT	 COMMUNICATIONS SCIENCE	Name: Le Son Tung..... Class/ Reg. No.: S208..... (12) Date: June 29 2026.....
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2026 SECONDARY 2 COMMUNICATION SCIENCE

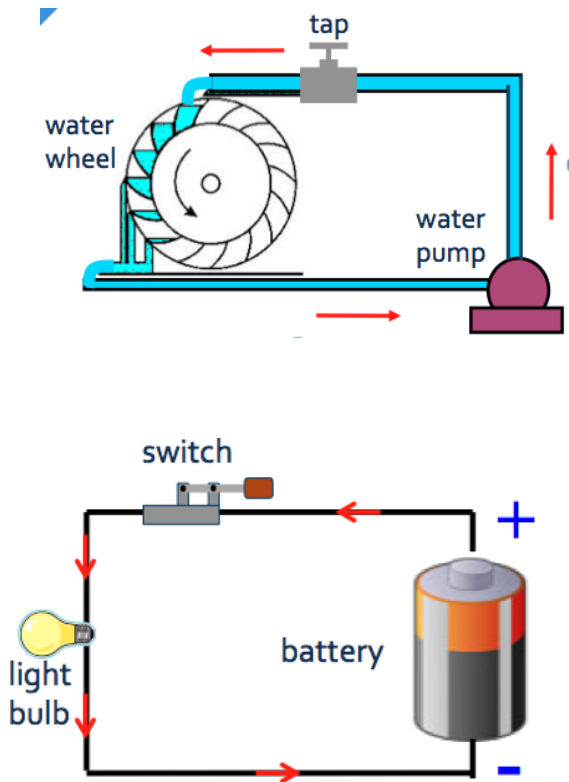
S/NO.: 1	TOPIC: Signals	UNIT: Current of Electricity	NOTES 1
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Knowledge	Cognitive Skills or Practical Skills
1.1 E.m.f., p.d., Current and resistance	(a) Describe e.m.f., potential difference, current, and resistance of an electrical system, stating their SI units.
1.2 Circuit diagrams	(b) Draw and interpret circuit diagrams and set up circuits containing electrical sources, switches, lamps, resistors (fixed and variable), ammeters and voltmeters, other electronic components (e.g. diode, LED, LDR, thermistor, capacitors, transistors).
1.3 Variable Resistances	(c) Investigate the effect of varying resistance on the current in the circuit using fixed or variable resistors (simple calculations of no more than 3 resistors arranged differently).
1.4 Combination of resistances	(d) Investigate how series or parallel arrangement of fixed resistors affects the current as an output of the system (Use of formula $V=IR$ is required).
1.5 Effect of currents	(e) Describe the applications of chemical, heating and magnetic effects of an electric current (include conversion of electrical energy to sound energy in speakers and conversion of sound to electrical energy using a microphone).
1.6 Measurement of current, voltage and resistance	Practical Skills (a) Set up simple electric circuits using wires and/or breadboard with resistors, electrical and electronic components and measure current and voltage using meters. (Graphs are NOT needed) (b) Demonstrate precision and accuracy in making measurements on ammeter and voltmeter (taking into consideration zero and parallax errors).
Essential Questions	1. (F) What is electricity? 2. (C) How does electricity operate within a closed circuit?

In our daily conversation, we use words such as electricity, voltage and current. In Science, there are specific terms and meanings for describing electricity. This section of the worksheet will provide you with the terms and definition used in electricity.

Electric Current

current only flows when the circuit is closed



Analogy:

Current flowing in wires is hard to imagine, therefore you may find it easier to relate it to water flowing in pipes.

The **pump** will make the water move through the pipe.

The **battery** makes the current go through the wires.

The **water** takes energy from the pump to the turbine.

The **electrons** carry energy from the battery to the motor.

- What is an electric current? definition: must memorise word for word
- An electric current, I , is the rate of flow of charges.

rate: how fast (concept of time)

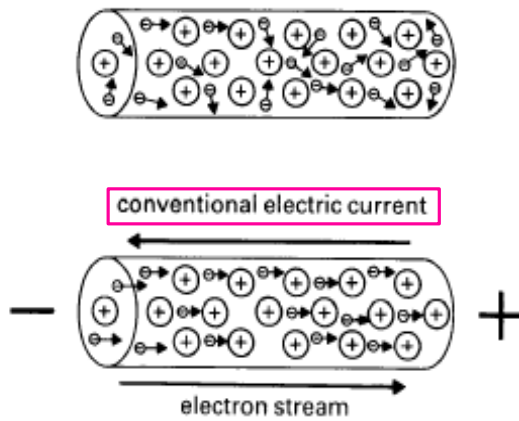
$$I = \frac{Q}{t}$$

Where I = current (in ampere, A)

Q = charges (in coulomb, C)

t = time (in seconds, s)

- The SI unit of current is the ampere (A).



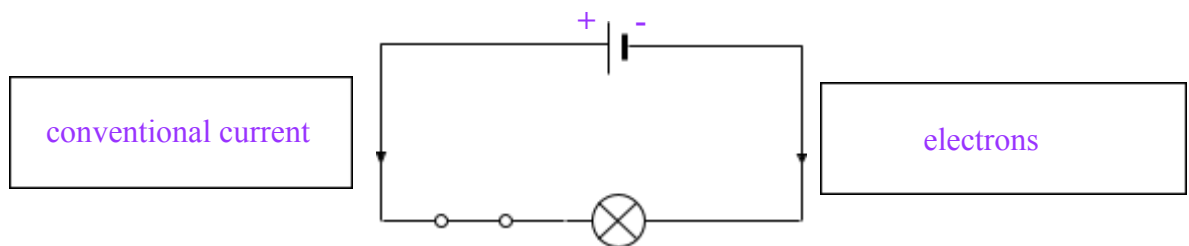
Question: How does current flow around the circuit?

Before connecting the battery across the wire, **free electrons** move about **randomly** in a wire.

When the battery is connected across the wire, a current flows in the wire. The free electrons being **negatively charged** will **move towards the positive terminal** of the battery. This is known as the electron flow.

Complete the diagram below, indicate the following:

1. Label the **+** and **-** terminals of the battery.
2. Indicate the direction of the **conventional current**. **flow from positive to negative**
3. Indicate the direction of the flow of electrons and label it as e.



- Although, it has been discovered that charge carrier in a conductor are **electrons** and these electrons **flow** from the **negative terminal** (—) to the **positive terminal** (+).
- For historical reasons, the direction of flow of **conventional current** is always taken to flow from the **positive terminal** (+) to the **negative terminal** (—).

**** Do keep in mind that we will always use the **direction of conventional current** for circuit analysis.**

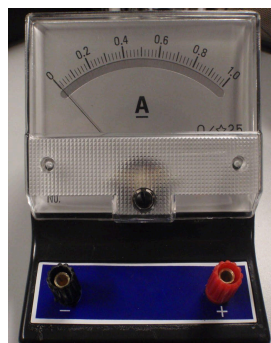
Measurement of current**Question.**

What is the SI unit of **current**?

Unit: ampere Symbol: A

Question.

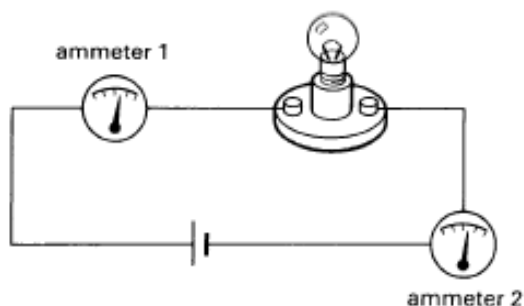
Which instrument do we use to measure current?



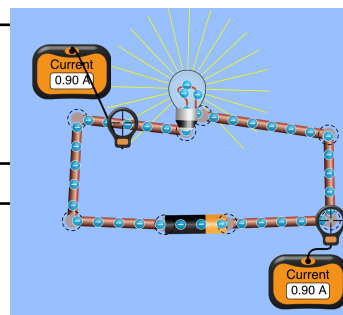
- An ammeter is used to measure current.
- An ammeter is placed in series along the wire and it will tell us how much current is flowing in the wire.

Example 1

Think about the readings on ammeter 1 and ammeter 2. What do you think it will show?

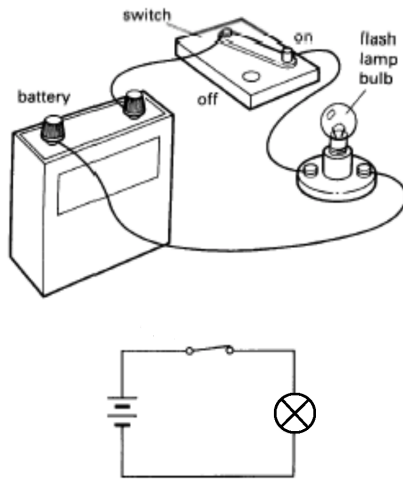


Scientific Thinking Routine	
Predict	Predict the values of current measured in ammeter 1 and 2. Share your answer to your peer partner.
	<u>I think it will be the same</u>
Observe	Observe the reading using the applet: https://phet.colorado.edu/en/simulation/circuit-construction-kit-dc-virtual-lab What did you observe?
Explain	Can you explain why?
	<u>There is a fixed amount of current flowing through the circuit</u>



Circuit Diagrams

A circuit diagram lets us represent a circuit on paper by using symbols for the various electrical components instead of drawing pictures.



Drawing the picture on the left would be difficult and time consuming.

Much easier is the circuit below drawn using recognized symbols.

The wires are shown by straight lines.

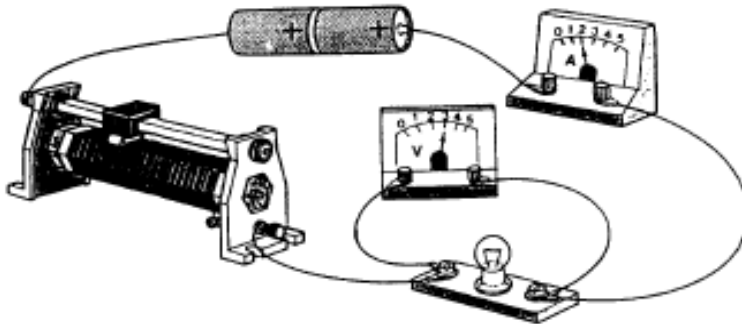
All the components have symbols, some of which look like the component, others do not.

The table below shows the various circuit symbols used to represent the components.

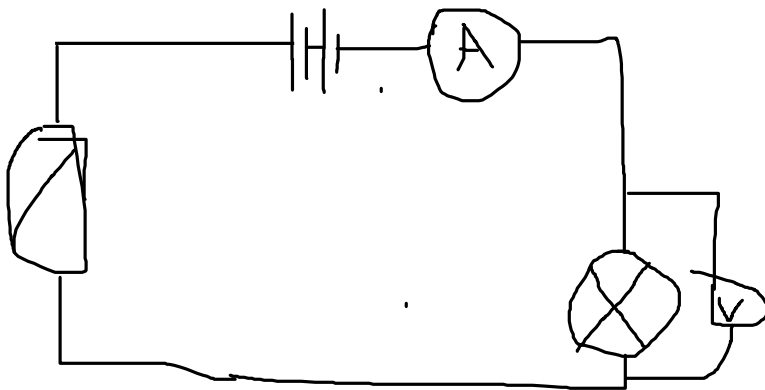
COMPONENT	CIRCUIT SYMBOL	COMPONENT	CIRCUIT SYMBOL
Cell		Resistors (fixed)	
Battery		Resistors (variable)	
Ammeter		Light Dependent Resistor (LDR)	
Voltmeter		Diode	
Joined wires		Light Emitting Diode (LED)	
Wires crossing (not connected)		Lamp	
Switch: Single pole single throw (SPST)		Transistor (BJT)	
Switch: Single pole double throw (SPDT)		Relay	
Switch: Double pole double throw (DPDT)		Thermistor	
Motor		Capacitor	

Example 2

Draw the circuit diagram for the arrangement shown below:



Circuit Diagram:



Computer Simulation:

Carry out a simulation of the circuit diagram that you have drawn above using the following circuit simulator:

https://phet.colorado.edu/sims/html/circuit-construction-kit-dc/latest/circuit-construction-kit-dc_en.html

Explore the functions of the circuit simulator.

What are the advantages and disadvantages of such a circuit simulator?

Advantage 1: *It is easier to understand*

Advantage 2: *It is easy to do*

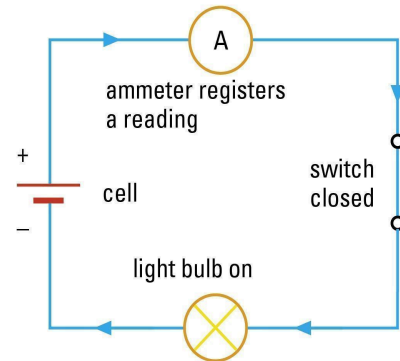
Disadvantage 1: *It is not realistic*

Disadvantage 2: *It is different from a real circuit*

After you have constructed your circuit, you may modify it to study the behavior of circuits if certain arrangements are made on page 7 (open circuit and short circuit).

Electric Circuit

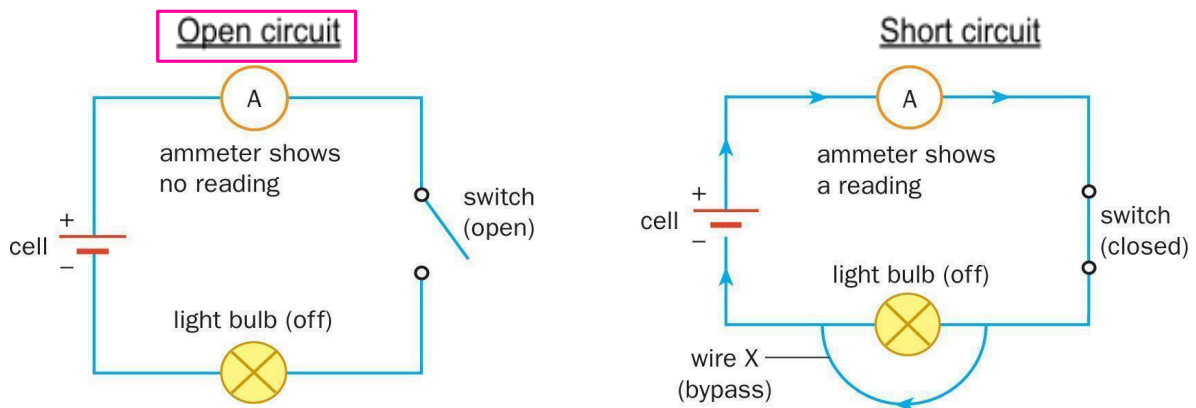
- An electric circuit is a closed-path (closed) through which charge can flow from one terminal of the electrical source to the other terminal.
- It consists of four main components:
 1. A source of electromotive force (e.m.f) that drives the electric charge around the circuit (e.g. a battery)
 2. A load on which moving charges can do a useful job (e.g. a bulb)
 3. Conductor (connecting wire) to connect the components together (e.g. copper wire), and
 4. Switches to open or close the circuit



Interpreting circuit diagrams

Use the following circuit simulator to create circuits shown below and observe the behavior:

https://phet.colorado.edu/sims/html/circuit-construction-kit-dc/latest/circuit-construction-kit-dc_en.html



Can you explain why it happens?

Open circuit occurs when there is a break in the circuit.

A break in the circuit means that current cannot flow through through it.

Besides an open switch, breaks in circuit can occur due to missing connections, loose wires, or a broken component.

Short circuit occurs when a part of the circuit connection is bypassed.

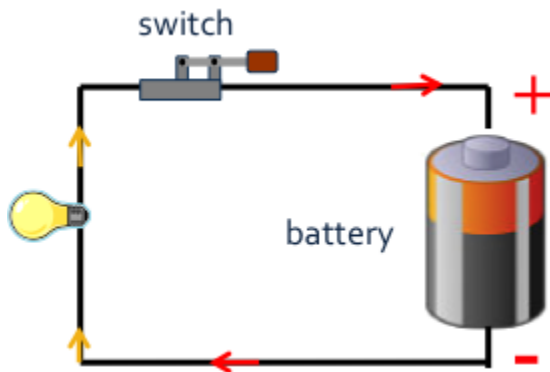
The circuit is closed but the bulb remains unlit.

This is because there is an alternative path of negligible resistance for current to flow through.

Therefore, the current does not flow through the bulb. In a short circuit connection, current flowing through the circuit can become very large; it would cause enormous heating and even fire.

Electromotive Force (e.m.f)

To cause the charge to flow in a circuit, we need energy. This energy is typically obtained from a source of e.m.f in a circuit. The battery produces an electromotive force which causes the charge to flow from the positive terminal to negative terminal.



EMF Source

A device which (battery or electric cell) transfers energy in the chemical potential store into other store(s) electrically.

positive charge

When a charge passes through a cell it gains energy from the cell.

This energy will later be dissipated in the rest of the circuit as the charge moves through the circuit.

Cell (or battery) : chemical potential store

Generator (or dynamo) : kinetic store

Thermocouple : Internal (Thermal) store

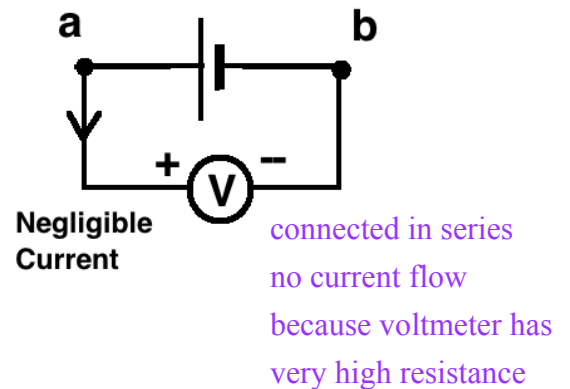
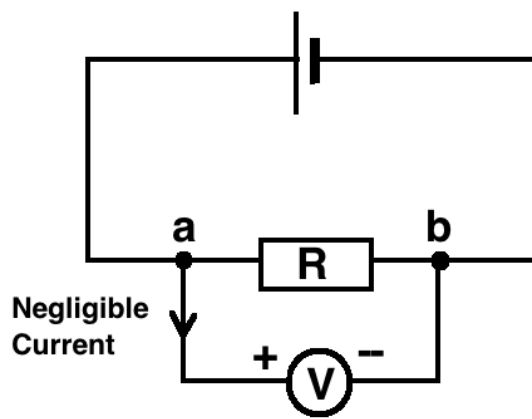
- **Definition:** Electromotive force is the work done per unit charge by the source in driving charges around a complete circuit.
- The SI unit of Electromotive Force is volt (V).

Potential Difference (p.d.)

- **Definition:** Potential Difference is the amount of work done per unit charge in driving charges through the component between the two points.
- The SI unit of Potential Difference is volt (V).

Measuring Potential Difference & Electromotive Force

A **voltmeter** is used to measure the e.m.f. of an electric cell and the p.d. across an electrical component.



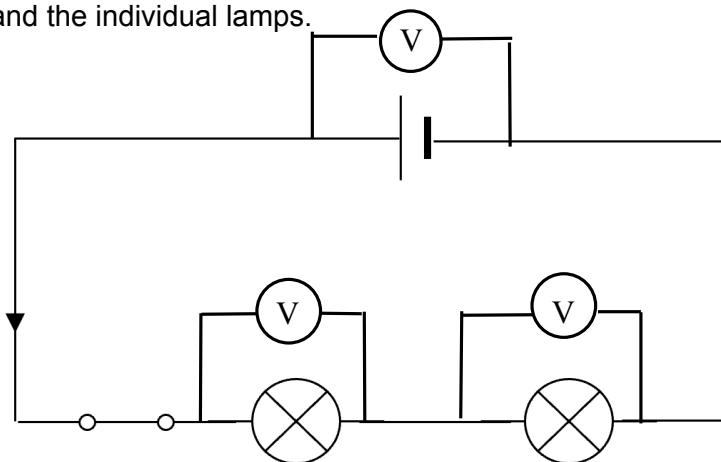
The circuit shows the set up for measuring the potential difference across the resistor R and the electromotive force across the cell.

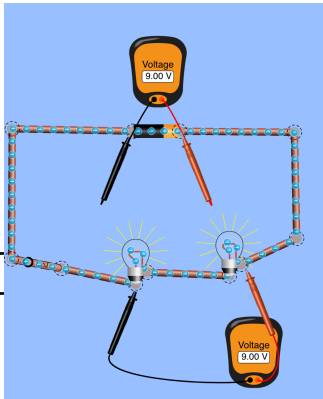
- Ideally, no current flows through a voltmeter. However, in practice, a small amount of current still flows through the voltmeter to move the needle. For our calculations, we will assume the ideal case that the current flowing through the voltmeter is zero.
- When measuring potential difference across R , the voltmeter must be connected parallel to the resistor and the circuit must be closed.
- When measuring the electromotive force, the voltmeter must be connected in series to the battery and not connected to anything else (the circuit must be open).
- **Try this experiment out to find the difference! Are the readings the same or different? Can you explain why?**

Example 3

What is the relationship between the p.d. of the battery and the light bulbs?

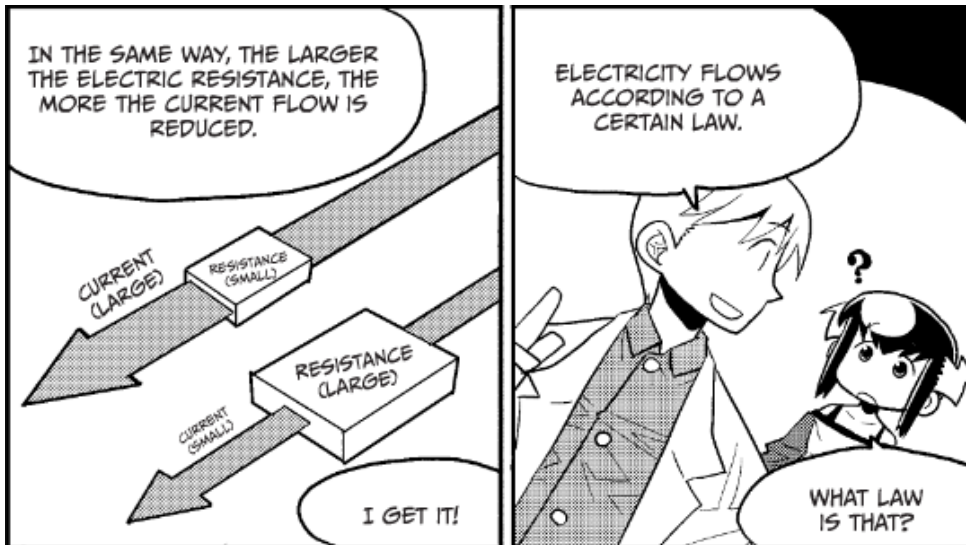
Using the circuit symbols, draw the **voltmeters** to measure the potential difference of the battery and the individual lamps.



Scientific Thinking Routine	
Predict	Predict the values of p.d. across the battery and the individual lamps. Share your answer to your peer partner.
	9V
Observe	Observe the reading using either an instrument or the applet: https://phet.colorado.edu/en/simulation/circuit-construction-kit-dc-virtual-lab What did you observe?
	
Explain	Can you explain why?
	$9V = 4.5V + 4.5V$

Resistance

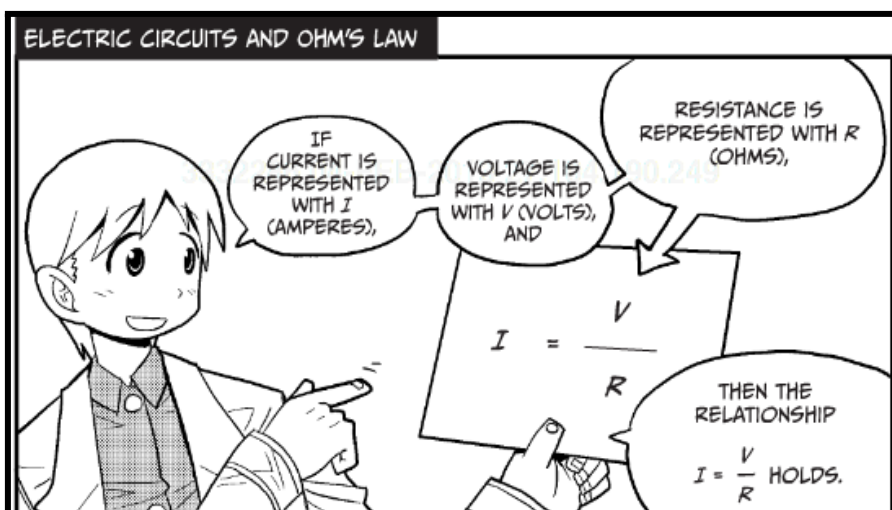
The resistance of a component is a measure of the opposition an electric current experience when it flows through the component.



- **Definition:** The resistance R of a component is defined as the

$$R = \frac{V}{I}$$

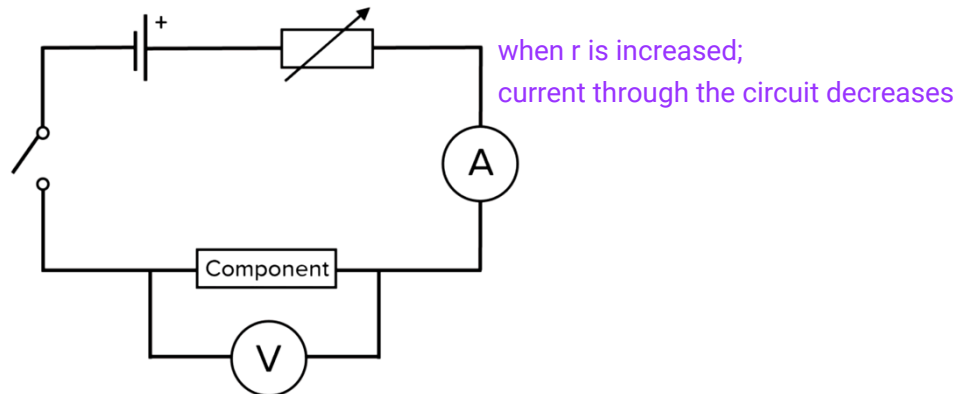
- Ratio of the potential difference across it to the current flowing through it.
 where $R = \frac{\text{resistance}}{\text{potential difference}}$ of component
 $V = \text{potential difference}$ across the component (in V)
 $I = \text{current}$ the component (in A)
- The S.I. unit of resistance is ohm (Ω)



I/V Characteristic Graph

Graph of I against V . I is y-axis, V is x-axis

Set up the circuit below and place the following items (resistor, filament lamp, Light emitting diode) in the position labelled as component.

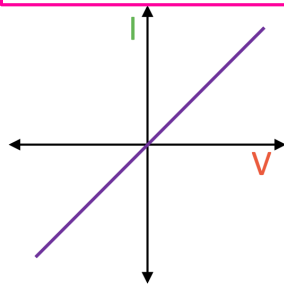


If we plot the current against the voltage (p.d.) for a component, we can see immediately whether it obeys Ohm's Law.

$$R = V/I$$

constant (no change) R

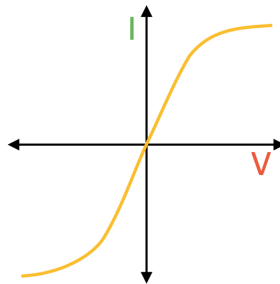
Ohmic Conductor



I is directly proportional to V

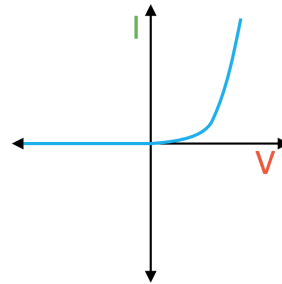
R changes

Filament Bulb



I is not directly proportional to V

Diode



R changes

Ohm's Law and Ohmic resistors

- Resistance of metallic conductors/resistors remains constant under steady physical conditions.
- Conductors that obey Ohm's Law are known as ohmic conductors.
- I-V characteristics of an ohmic conductor. A straight-line graph shows that the gradient (i.e. the inverse of resistance R of the conductor) is a constant, hence R is also a constant.

Non-ohmic resistors

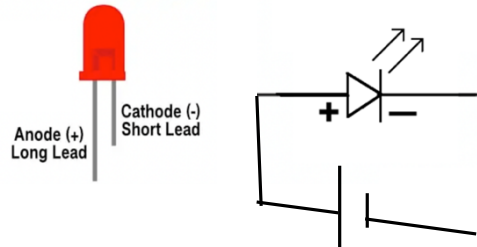
- They **do not obey** Ohm's Law and their resistance R can vary.
- Their I-V graphs are not straight line which means the ratio V/I is not a constant.

Filament lamp

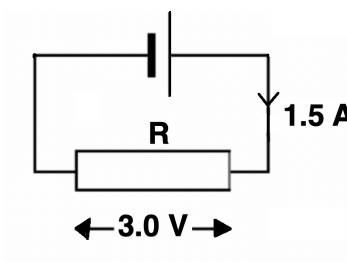
As the p.d. across the lamp increases, the current does not increase proportionally.

Light Emitting Diode (LED)

As the p.d. across the lamp increases, the current does not increase proportionally. However, the diode must be connected correctly as shown in the diagram below.

**Example 4**

(a) What is the value of the given resistor R?

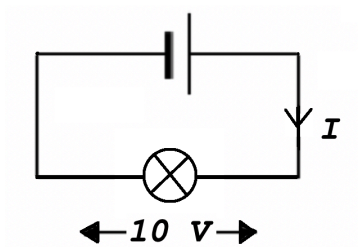


$$R = \frac{V}{I}$$

$$R = \frac{3.0}{1.5}$$

$$= 2.0 \, \Omega$$

(b) What is the current flowing through the following circuit if the bulb has a resistance of $50 \, \Omega$?



$$50 \, \Omega = \frac{10V}{I}$$

$$I = \frac{10V}{50\Omega}$$

$$= 0.20 \, \text{A (3s.f.)}$$

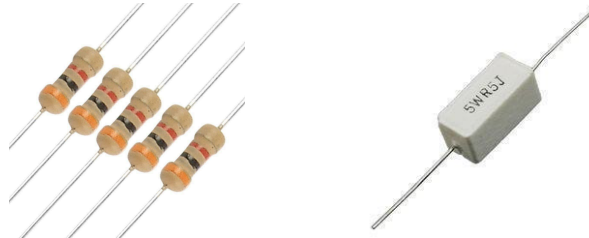
What are resistors?

- A resistor is a conductor in a circuit that has a known value of resistance.
- They are mainly used to control the size of current flow flowing in a circuit.
- There are two types of resistors — fixed and variable (or rheostats).



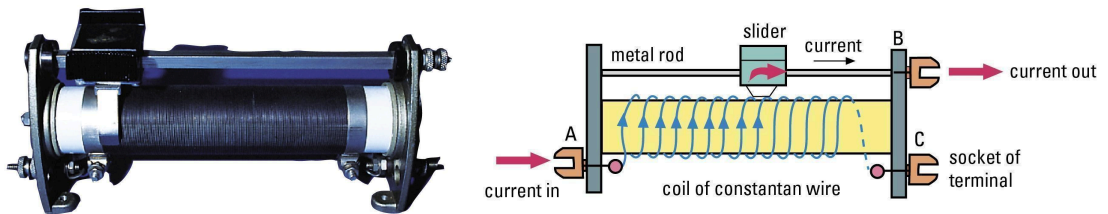
Fixed resistors

- They have constant value of resistance.
- Examples include carbon film resistors, carbon composition resistors, tin oxide resistors and wire-wound resistors.



Variable resistors (rheostats)

- The resistance of a variable resistor can be changed.
- A rheostat is included in a circuit to vary the size of current flowing through it.

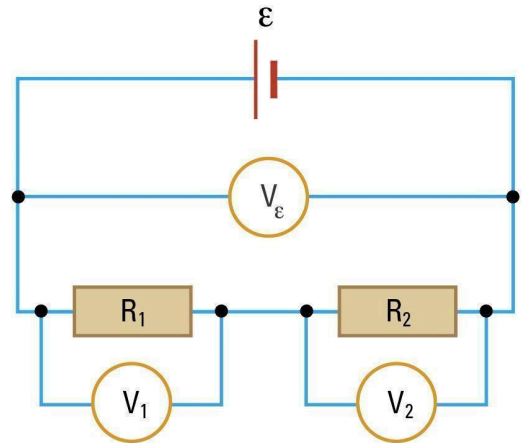
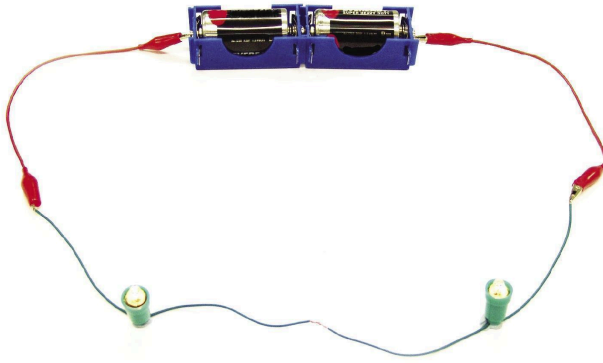


Series and Parallel Circuits

Series Circuits

Current in a series circuit

- Circuits in which all of the components are placed so that there is only one path for the current to go around are called series circuits.



Potential difference across a series circuit

- In a series circuit, the sum of the potential differences across each component is equal to the potential difference across the circuit.

$$\Sigma = V_1 + V_2$$

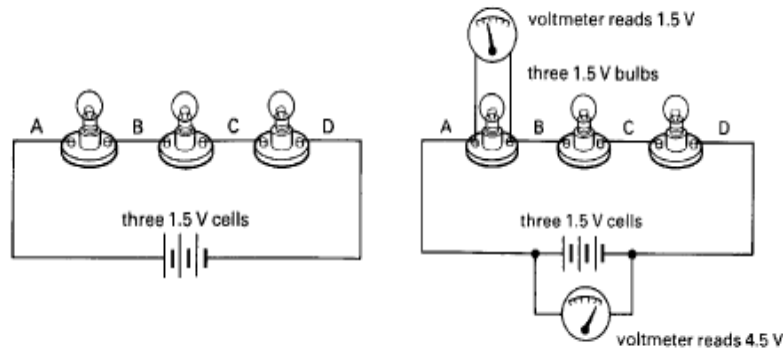
- Assumption: Even if the cell has internal resistance, its terminal p.d. V_E would be equal to $V_1 + V_2$.

the cell has

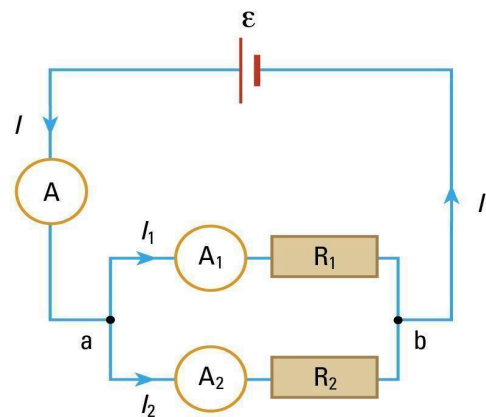
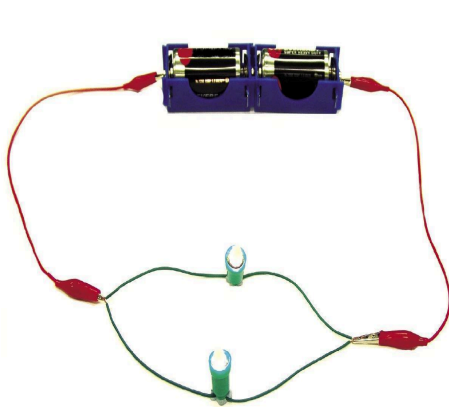
Assume that the battery has zero internal resistance.

Characteristics of series circuits

- The current flowing through each bulb is the same.
- The sum of the potential difference across the bulbs is equal to the e.m.f of the battery.
- The light bulbs shine with the same brightness. (does not imply that bulbs are very bright)

**Parallel Circuits****Current in a parallel circuit**

- In a parallel arrangement, there is more than one path through which current can flow.
- Circuits in which the components are placed so that there are two or more paths for the current to go around are called parallel circuits.



Question: Explain why Ammeters A_1 and A_2 show smaller current readings compared to ammeter A .

The current I splits into I_1 and I_2 at junction a and recombine into I at junction b . The sum of the currents measured by ammeters A_1 and A_2 is equal to the current measured by ammeter A .

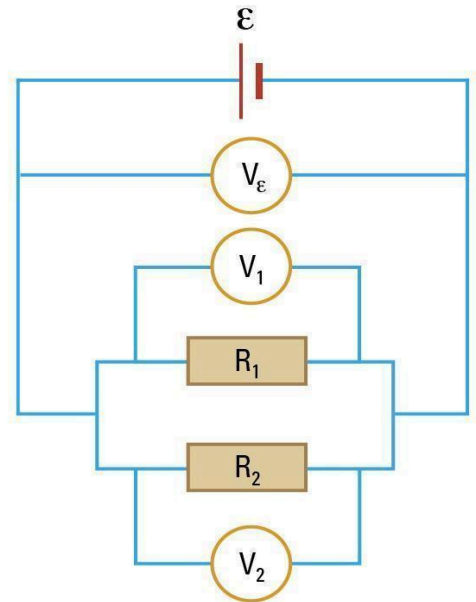
$$\text{total } I = I_1 + I_2$$

Thus,



Potential difference across a parallel circuit

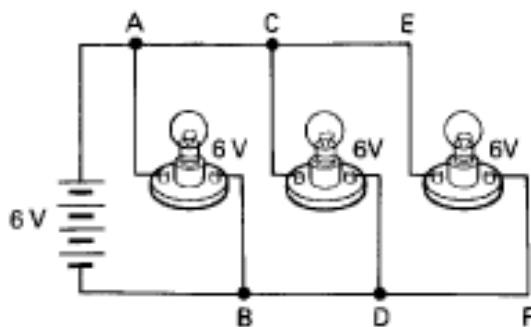
- All the **voltmeters** have the **same reading**.
- The potential difference across each of the two resistors in parallel is the same as the potential difference across the entire circuit, V_ϵ .
- If the cell has negligible internal resistance, the p.d across each resistor would be equivalent to V_ϵ



$$V_\epsilon = V_1 = V_2$$

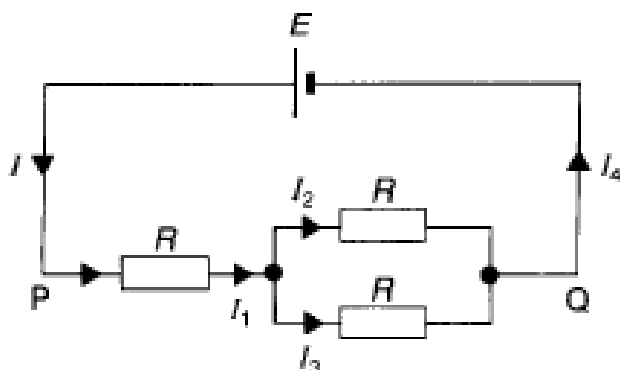
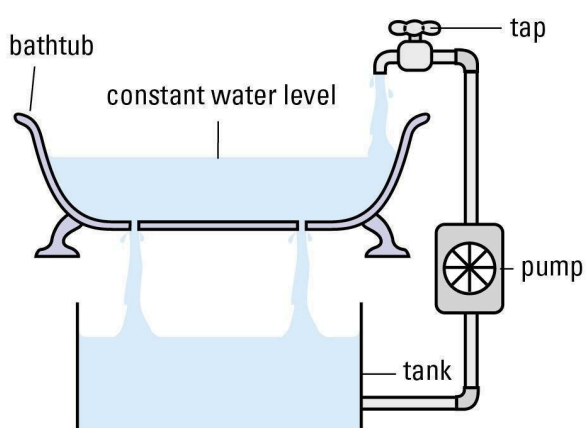
Characteristics of parallel circuits

- The potential difference across each bulb equals the e.m.f of the battery.
- The **sum of the currents in the bulbs** equals the **current passing through the battery**.
-



Combination circuits

Most circuits will contain some components in series and some in parallel.

**Analogy:**

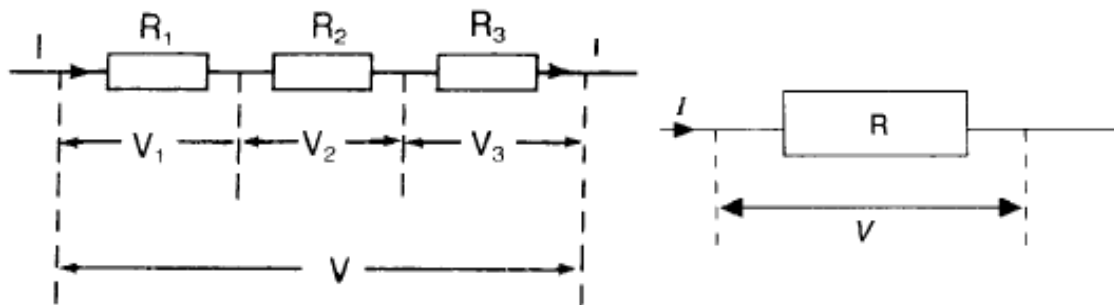
Effective Resistance

total resistance of the circuit

When we have two or more resistors in a circuit, we can see what the combined effect of their presence is. This is known as the effective resistance of the resistors.

Resistors in Series

- The following three resistors have a current I passing through them.



This circuit is equivalent to...

...this one.

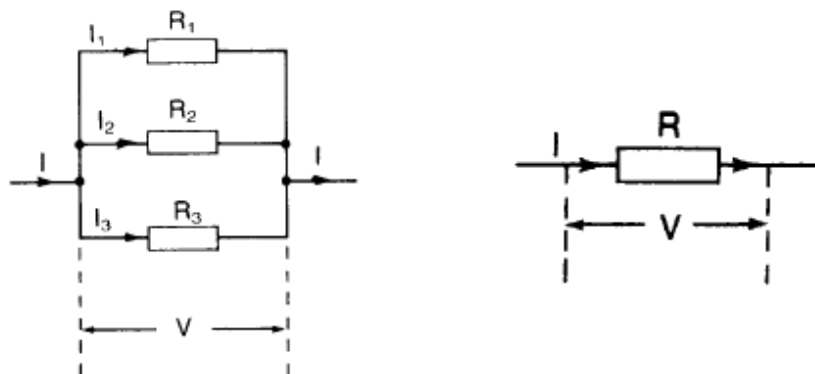
Thus, the effective resistance of R_1 and R_2 and R_3 connected in **series** is R .

In general,

$$R_{\text{effective}} = R_1 + R_2 + R_3$$

Resistors in Parallel

- The following three resistors have a potential difference V across them.



This circuit is equivalent to...

...this one.

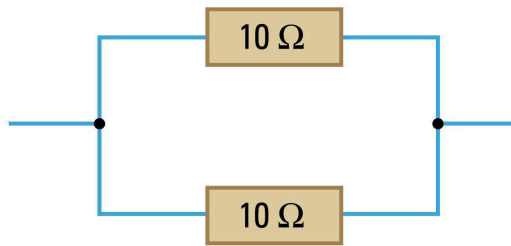
Thus the effective resistance of R_1 and R_2 and R_3 connected in parallel is R .

In general,

$$\frac{I}{R_{\text{eff}}} = \frac{I}{R_1} + \frac{I}{R_2} + \frac{I}{R_3}$$

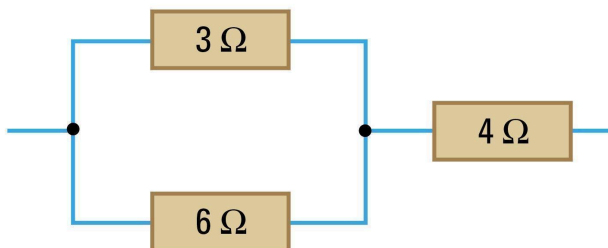
Example 5: Find the net resistance in the following arrangements:

(a)



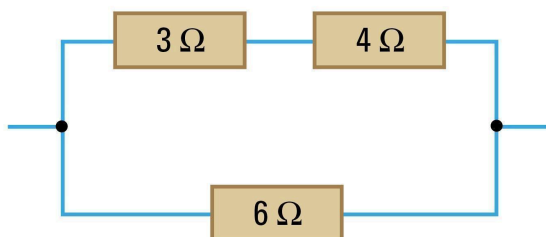
Working:

(b)



Working:

(c)

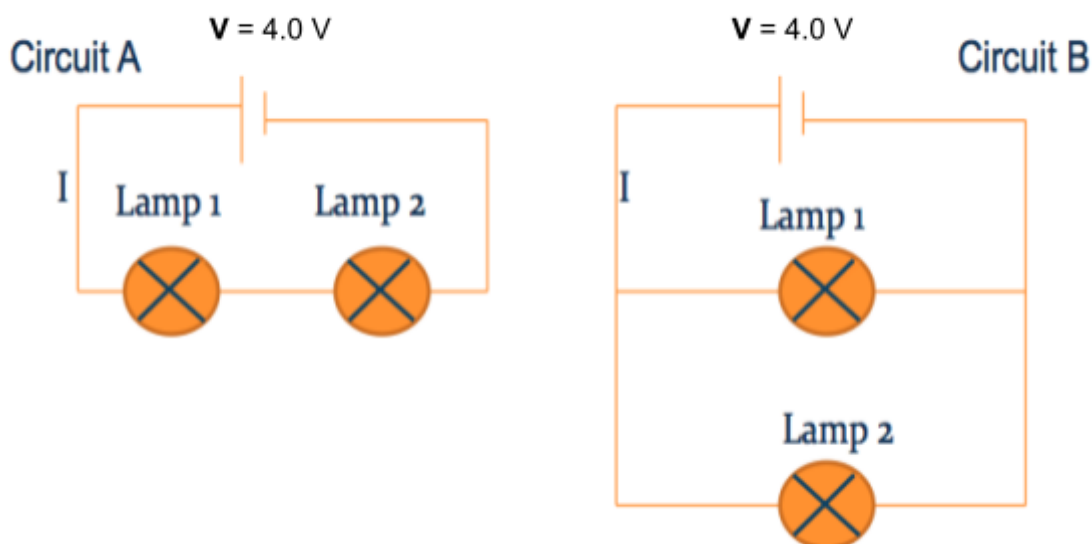


Working:

Connecting light bulbs in series and parallel

The diagram below shows a series and a parallel connection of two lamps.

(Assume that the resistance of each lamp is the same)

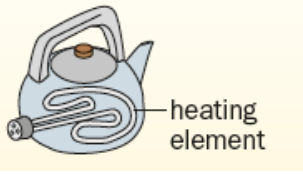
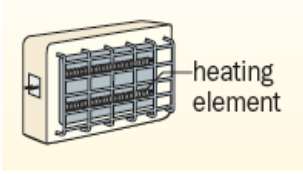
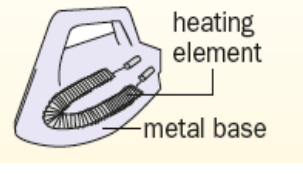
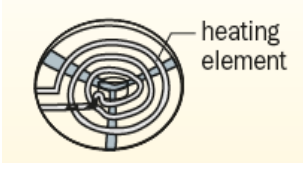

Scientific Thinking Routine

Predict	Predict which lamp(s) is/are brighter and which ones are dimmer. Share your answer to your peer.
Observe	Observe the reading using the applet: https://phet.colorado.edu/en/simulation/circuit-construction-kit-d-c-virtual-lab What did you observe?
Explain	Can you explain why?

Heating and Magnetic effects of current

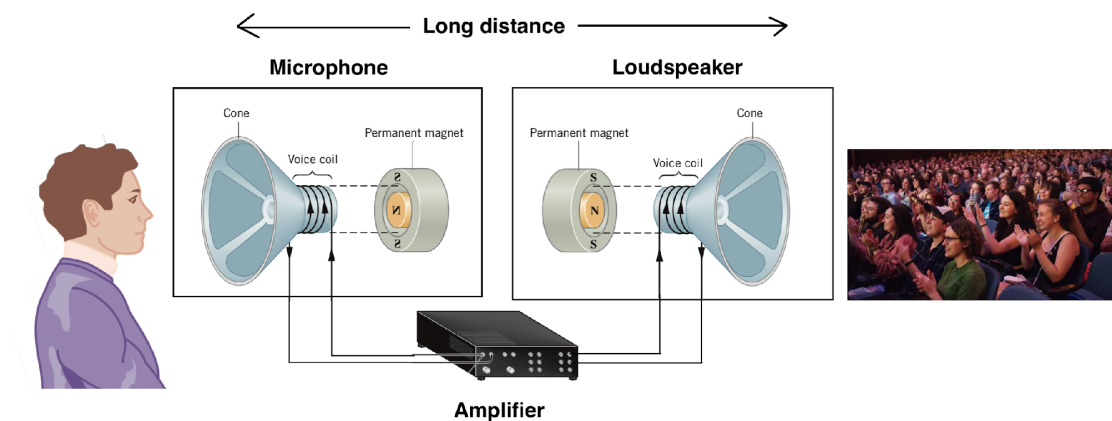
Electric Heating

- Some household appliances have **heating elements** that transfers electrical energy to thermal energy.
- Heating elements are usually made of Nichrome because it has high **resistivity** and can withstand **high temperature**.
- When an **electric current** passes through the heating element, it heats up rapidly.
- Examples of electrical appliances with heating elements include:

Electrical Appliance	Name	Electrical Appliance	Name
	Electric Kettle		Electric Radiator
	Electric Iron		Electric Hotplate

Magnetic effect of current

- The **microphone** is a useful device that transfers **sound to electricity**.
- The **loudspeaker**, on the other hand, transfers **electricity to sound**.
- This mechanism allows a large audience to be addressed without much effort by the speaker.
- The sound gets transferred into electricity by means of **vibration of a cone in a strong magnetic field**, which causes an electrical signal to be generated.
- The signal is then **amplified** and sent to **another cone in the presence of a strong magnetic field**. This causes the cone to vibrate to and fro and generates a **loud sound**.



THE END